

A 36 V Programmable Instrumentation Amplifier With Sub-20 μV Offset and a CMRR in Excess of 120 dB at All Gain Settings

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Abstract—A 36 V capable programmable gain instrumentation amplifier (PGA) is presented with sub-20 μV offset, sub-0.2 $\mu\text{V}/^\circ\text{C}$ offset drift and a common-mode rejection (CMRR) that exceeds 120 dB at all gain settings without any trimming. It is the first 36 V capable precision PGA implemented in a high-voltage CMOS process, which, in addition, incorporates several additional functions, such as the detection of input and output fault conditions, provisions for improving system-level settling time and an input switch network. All op-amps used in the PGA employ chopper stabilization with a notch filter that removes chopping glitches, leading to low offset and drift and no $1/f$ noise. The PGA has a total of 22 gain steps (binary steps between 1/8 to 128, each with an optional multiplying factor of 1 or 1.375) with better than 0.1% gain accuracy, <0.001% nonlinearity and sub-2 ppm/C gain drift. The input switch network, in addition to acting as a 2-channel multiplexer, also enables various system-level diagnostic features. The PGA is implemented in a 0.35 μm CMOS process with a 36 V extension, has a 3.6×2.4 mm chip area and consumes a total quiescent current of 3 mA.

Index Terms—Chopper, CMRR, high-voltage CMOS design, industrial instrumentation, instrumentation amplifier, programmable gain.

I. INTRODUCTION

WHILE several application-specific sensor-interface ICs [1] and integrated smart sensors [2] have been published and made available commercially in recent years, in industrial signal-acquisition applications there is still a strong need for a more universal interface solution for various signal sources, supporting an effective resolution of 16 bit and above. The system implementation of such interface solutions usually includes a programmable or fixed gain instrumentation amplifier (IA) followed by a high-resolution A/D converter (ADC). This paper focuses on the programmable-gain instrumentation amplifier (PGA) implementation.

In industrial signal-acquisition applications, apart from sensor outputs of a few millivolts, larger signals of up to ± 10 V exist, such as those coming from remote transmitter outputs. Signals in this wide range need to be extracted and translated without corruption to the low-voltage domain in which most A/D converters and microcontrollers operate. Moreover, these

TABLE I
KEY DESIGN GOALS AND SYSTEM REQUIREMENTS

Parameter	Requirement
Full Scale Input Range	$\pm 20\text{mV}$ to $\pm 10\text{V}$
Full Scale Output Range	up to $\pm 5\text{V}$
Input Impedance	$> 10\text{M}\Omega$
Common Mode Rejection	$> 100\text{dB}$
Common Mode Input Range	$\pm 16\text{V}$
Output Referred Offset Voltage	$< 3\text{mV}$
Output Referred Offset Voltage Drift	$< 1\mu\text{V}/^\circ\text{C}$
Gain Error Drift	$< 4\text{ppm}/^\circ\text{C}$
Linearity	$< 0.002\%$
Output Referred Noise (0.1 to 10Hz)	$< 62\mu\text{Vpp}$
On Chip MUX and Diagnostics	YES

signals can have up to ± 10 V of common-mode voltage due to long connection lines and different ground potentials, thus requiring a common-mode rejection-ratio (CMRR) in excess of 100 dB up to 100 Hz. Due to the variety of signal sources, high input impedance ($> 10\text{M}\Omega$) is essential, making systems based on resistive input dividers or on capacitive input sampling impractical, due to their low dynamic impedance and input current draw. To limit the need for system calibration and to maintain the accuracy over time and temperature, offset drift, gain drift and nonlinearity need to be minimized, while $1/f$ noise needs to be eliminated. Table I summarizes the key design goals and system requirements.

The PGAs and IAs currently available for use in industrial signal-acquisition applications can generally be divided into two categories. A first category is implemented in high-voltage (HV) bipolar technology, and use laser-trimming to achieve good noise and DC precision [3], [4]. These are capable of handling input common-mode and differential voltages of up to ± 15 V, which are still commonplace in industrial applications. The drawbacks of such ICs include limited programmability, lack of diagnostics and auxiliary functions, and relatively high cost due to the use of bipolar technology and trimming. At low gain settings, most of these ICs also have a limited CMRR of about 80 dB.

A second category of PGAs is implemented in analog CMOS technology [5], [6]. This enables the use of dynamic offset cancellation techniques to improve DC precision, thus eliminating the need for trimming. Moreover, the use of CMOS technology allows for the integration of a digital interface, additional functionality such as sensor excitation/linearization and the ability to detect and indicate internal or system level fault conditions.

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Since this can make the overall readout system much more robust against errors, there has been a considerable interest in PGAs with such additional functionality. However, until now, the relatively low supply voltages of such CMOS PGAs has made them unsuitable for many industrial applications, where an input range of ± 15 V is required.

In this paper, we present a precision PGA implemented in a HV-CMOS technology that covers this wide input range and integrates additional digital control and diagnostics functions while meeting or exceeding the DC specifications of similar bipolar HV IAs and PGAs.

This paper is organized as follows. Section II presents the system-level design of the PGA and discusses the architecture, the gain network and the design approach. Section III outlines the opamps and other key circuit blocks that are critical to the overall performance of the PGA. In Section IV, the additional functionality that the use of CMOS technology allows, such as error detection circuitry, is described. In Section V, measurement results are shown. Finally, conclusions are presented in Section VI.

II. SYSTEM-LEVEL DESIGN

A. Overview

This section gives a quick overview of the chosen architecture while further details are provided in subsequent sections. The key design goals that mandated the architectural choices are listed in Table I. The choice of a high-voltage-capable CMOS process enabled implementation of the on chip MUX, digital control and diagnostics features. A current-mode IA structure was selected for its high input impedance, wide common-mode input range and high CMRR capability. The gain range based on the full scale (FS) input- and output-range requirements was chosen to be 1/8 to 128 in binary steps. In order to minimize the unused output range between gain steps an intermediate multiplying factor of 1 or 1.375 was added. The overall PGA contains two input opamps, two output opamps, and two precision current mirrors each employing two OTAs. To meet the offset, offset drift and low frequency noise requirements, all of these amplifiers employ chopper stabilization. Chopping glitches could potentially cause settling and aliasing problems in successive parts of the signal chain, especially at high gain settings. To avoid this, chopping glitches were filtered out by using the notch-filter approach introduced in [7]. The chopping frequency of 250 KHz was chosen to be high enough to keep the capacitors in the notch-filter small but low enough to avoid large input bias currents and unnecessary amplifier bandwidth requirements.

B. IA Architecture

This sub-section gives an overview of the instrumentation amplifier architectures considered and provides details about the architecture used in this PGA. Capacitive sampling techniques such as [8] can achieve very good noise and DC performance characteristics. However, they provide no continuous signal path to the output and thus may suffer from sub-sampling and aliasing effects. Furthermore, achieving an input impedance

in the order of 10 M Ω at a reasonable sampling frequency, would require an impractically small capacitance, whose high kT/C noise would limit SNR. Similarly, the low input impedance of amplifiers with input resistor structures (such as a divider) is unacceptable in our application.

Recently published Current-Feedback IA architectures [9], [15], [16] rely on transconductors to convert both the input and feedback voltages into currents, which are then balanced at a virtual ground node. Although such amplifier's can achieve high CMRR and low offset, their input range is restricted to tens of millivolts. Due to transconductor mismatch, they also suffer from limited gain accuracy. Although this can be increased by using transconductance-setting resistors and local feedback loops [9], [15], this is at the expense of increased input-referred noise.

One of the most commonly used IA topologies in the industry is the 3-opamp topology [3], [4], [10]. Having dedicated opamps at each input provides a high input impedance for any input signal within the opamp's common-mode input range, along with a continuous signal path to the output. The gain is usually set at the front-end for best noise, DC and AC performance. Using precision chopper stabilized opamps in such an IA (Section III-A) makes it possible to meet the offset, offset drift and low frequency noise requirements. This topology was, therefore, chosen as the starting point of the PGA's architecture. However, the standard 3-opamp structure has two important drawbacks. Firstly, the use of voltage gain in the input amplifiers reduces the input common-mode voltage range for high gains, since the outputs of the input amplifiers can clip. Secondly, the CMRR is determined by the matching between the resistors around the output amplifier, which is typically limited to about 80 dB at low gains.

Another current-mode instrumentation amplifier approach, described in [11]–[14] can provide wide common-mode input range and high CMRR while maintaining the high input impedance, noise and offset/drift advantages of the 3-opamp topology. This type of current-mode IA structure is implemented in this design as shown in Fig. 1. The input amplifiers A_{I1} and A_{I2} induce the input voltage across the programmable resistor network. The current through this resistor network, which also flows through the output stage of the input amplifiers, is mirrored by precision bi-directional current mirrors and fed into the summing nodes of the output opamps A_{O1} and A_{O2} , where it is converted into a differential output voltage via the feedback resistors R_{O1} and R_{O2} . The total PGA gain $G = A_{CM} \cdot G_{IN} \cdot (R_{O1} + R_{O2})$ where A_{CM} is the fixed gain implemented in the current mirror and G_{IN} is the transconductance set by the resistor network. Through the use of this V-I and I-V conversion, the input common-mode range becomes independent of the gain, and the CMRR does not depend mainly on resistor matching, but on the mismatch of output resistance, open loop gain and CMRR of the individual input amplifiers [13]. Equally important is that this architecture is compatible with the high-voltage/low-voltage (HV/LV) partitioning method described in Section II-D, leading to a more efficient and compact design. The only disadvantage of the approach is that in current-mode systems certain internal nodes

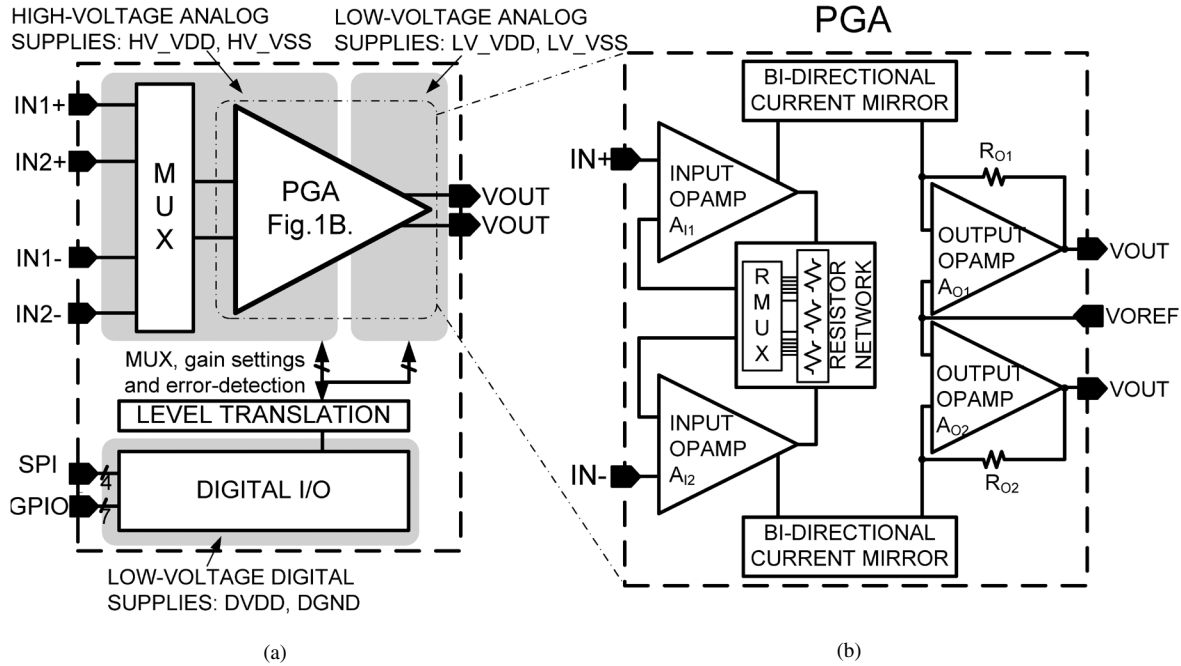


Fig. 1. Block diagram of the presented PGA: (a) the key system-level components and supply domains and (b) a detailed block diagram of the signal path.

need to have a high impedance to assure low distortion, which is difficult to achieve over frequency because of the necessary HV cascoding transistors.

Depending on the selected gain, the precision of the input and output amplifiers and the current mirror affects the overall performance of the PGA to a different degree. While the output error contribution of the input amplifiers is multiplied by the PGA gain, the contribution from current mirror and output amplifiers is only modified slightly by the chosen output scaling factor of 1 and 1.375. Therefore, to meet the offset, offset drift, CMRR, PSRR and low frequency noise requirements at high as well as low gain settings, each of the opamps was chopper stabilized.

The DC precision of the current mirrors is critical for the overall offset, gain accuracy, drift performance and linearity, especially in lower gain and attenuation modes. The static matching of a regular or an impedance-boosted current mirror yields insufficient DC precision. A potential solution would be to use dynamic element matching, where the input and output transistors are exchanged to average out their mismatch. However, the transistor mismatches would lead to large current glitches at the mirror output, which would subsequently be converted into large voltage glitches by the output amplifiers.

Instead, a resistive current mirror solution was implemented as shown in Fig. 2. Referring to the P-side current mirror, the input current I_G flows through RP1 and the feedback loop consisting of OTA A_{CM1} and cascode transistor MPH2 ensures that the voltage across RP2 matches that across RP1, resulting in the mirror output current $I_{OUT} = I_G \cdot RP1/RP2$. The precision of this circuit depends on the accuracy of the OTA and the mismatch of the resistors. The resistors are implemented using a thin-film layer available on the process, and have sufficient static matching when proper precision layout techniques are used. It is

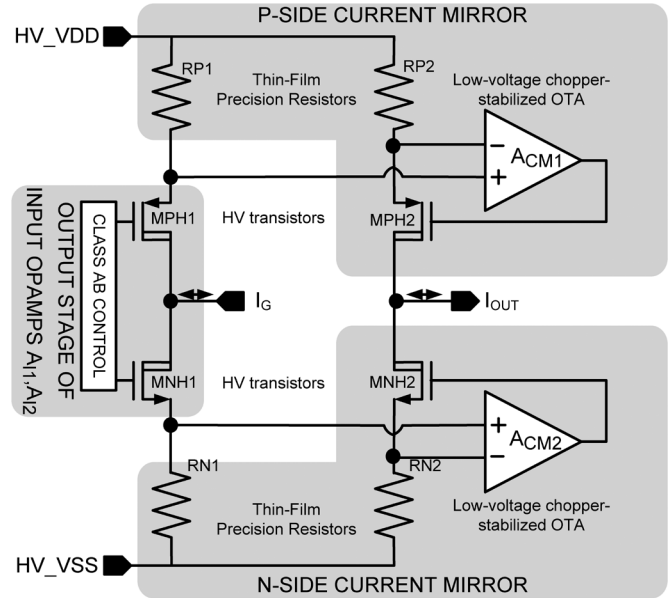


Fig. 2. The implementation of the bi-directional current mirrors from Fig. 1.

important to note that while the signal gain of the current mirror is set by the RP1/RP2 (and RN1/RN2) ratio, the OTA's noise and offset contribution is scaled by the ratio of the output resistor (R_{O1} or R_{O2} in Fig. 1) to the mirror resistor RP2 (RN2). Therefore, the value of the mirror resistors must be chosen, on the one hand, to minimize the voltage drop across them to avoid input range restrictions and, on the other hand, to minimize the contribution of errors from the current mirror amplifiers, such as offset and noise, to the output. This so-called noise gain is equal to 5 in this design. The OTA's offset, offset drift and 1/f noise are minimized by the same chopper stabilization scheme

used for the input and output opamps, and which is described in Section III-A.

C. Gain Network

To cover full-scale input ranges from $\pm 10\text{ mV}$ to $\pm 12\text{ V}$, the gain range was selected to be 0.125 V/V – 128 V/V in binary steps. These binary steps are implemented by changing the transconductance of the front-end comprising amplifiers A_{I1} , A_{I2} and the resistor network. In addition, to maximize the signal-to-noise ratio at the output of the PGA, and therefore relax the requirements of the ADC that typically follows the PGA in measurement systems, an additional 1 V/V or 1.375 V/V gain factor was realized by scaling the feedback resistors R_{O1} , R_{O2} .

Programmable transconductance stages typically rely on either an R-2R type resistor network (Fig. 3(a)) or on a tapped resistor string (Fig. 3(b)) to set the transconductance I_G/V_{IN} . When a large number of transconductance steps are required, the R-2R ladder is compact but, since the input signal voltage is amplified and the amplifier will have an output swing limitation, places severe limitations on the common-mode input range (CMIR). This approach also limits the signal bandwidth according to the gain-bandwidth relationship typically associated with voltage feedback. The tapped resistor string approach can have a much improved CMIR performance if the unused resistors to the left of the selected tap point switches in use are shorted out as shown in Fig. 3(b). In this case the voltage gain to the output of the amplifier is always 1 and the transconductance is changed by the total impedance to the right of the selected switches. The signal bandwidth is then independent of the transconductance setting, but the number of resistor segments goes up exponentially with the number of transconductance settings. This costs chip area and complicates layout and metal matching without adding significantly to system accuracy.

This PGA uses a combination of the two approaches to optimize the input range, dynamic performance, resistor area and gain error characteristics. This is illustrated in Fig. 4, with the dotted lines indicating the $G = 128$ configuration. For higher gains the R-2R type resistor network is used while the lower gains are implemented with a tapped resistor network. By doing this, the CMIR limitations are reduced as the voltage gain to the output of the amplifier is reduced (provided that the series resistors to the left of the tap point $G = 4$ are shorted as shown). In addition, the arrangement lends itself to easier layout and better matching performance at the more critical high gain settings. For lower gain settings the series resistance approach is used, therefore the CMIR is not limited and the bandwidth stays constant with changing gains in this region. The total number of resistor segments for the network shown (11 steps) is 94 compared to the 1024 that would be required if the whole network was implemented as a series resistor string. This means significantly less area, along with less layout complexity and the opportunity of better optimization, better matching and less series (parasitic) metal resistance. Most importantly, an additional degree of freedom is obtained, that can be used to optimize the gain network for system common-mode input range, bandwidth and chip area requirements.

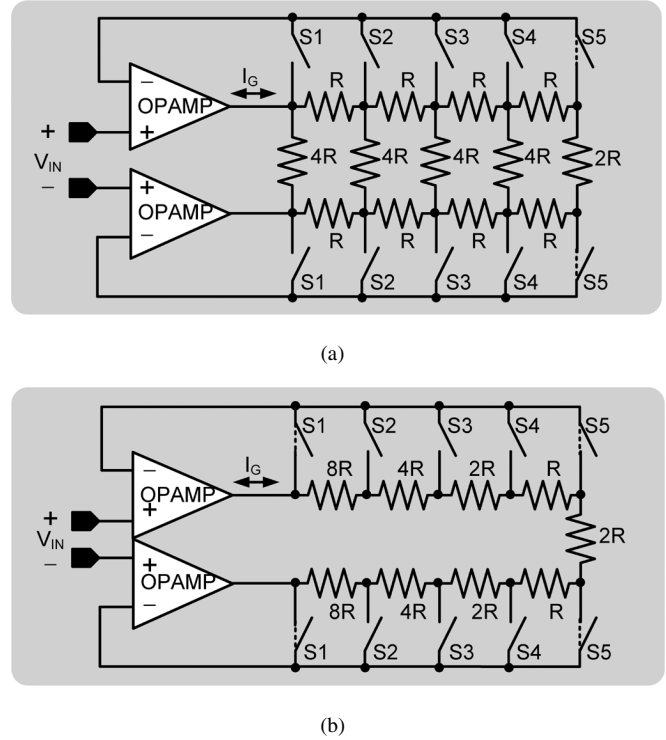


Fig. 3. Commonly used transconductance resistor networks based on (a) an R-2R ladder and (b) the addition of series resistors.

D. High-Voltage Design

To be able to handle the various industrial signal levels, the primary design goal for the PGA is to be able to operate from a $\pm 18\text{ V}$ supply, with an input common-mode range of $\pm 16\text{ V}$. The output amplifiers operate at a separate supply voltage of 3 to 5 V , which is essential to be able to drive a low-voltage ADC. This output supply voltage can be freely chosen to be anywhere within the high-voltage supply range. Similarly, the digital support circuitry also operates at a supply voltage of 3 to 5 V that can be freely chosen within the high voltage supply range. As a result, several analog and digital signal transfers between these various supply domains on the chip are required.

In order to meet the input range requirements along with the required digital gate density, a CMOS process is used that has 5 V thin-oxide transistors, thin-oxide transistors with extended drains for 36 V operation between drain and source, and thick-oxide gate transistors capable of handling 36 V across both gate-source and drain-source. Unfortunately HV-capable transistors have degraded performance and matching, which makes them unsuitable for use in gain stages of amplifiers and other parts of the PGA where good analog performance is required. However, the process allows the better performing low-voltage CMOS transistors to be isolated from the substrate. Therefore, throughout the design, the approach was to use isolated circuit blocks, with low-voltage transistors in the signal path, which are driven by on-chip voltage regulators. This is illustrated in Fig. 5. In addition to the three external (user) supply domains labeled with prefixes HV_ (high voltage), LV_ (analog low voltage) and D (digital), there are four internally generated supply domains. The clock conditioning and chopping switches at the input of

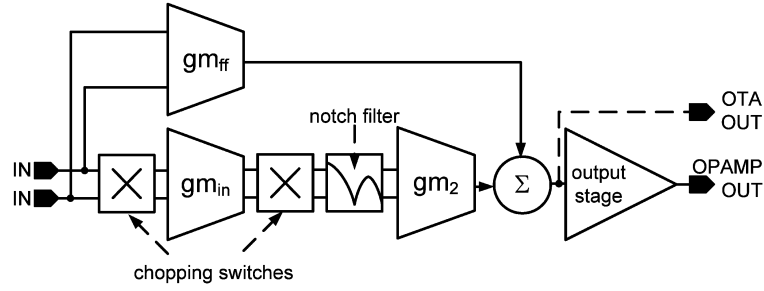


Fig. 6. Chopper stabilized amplifier structure used in the input and output amplifiers and current mirror OTAs.

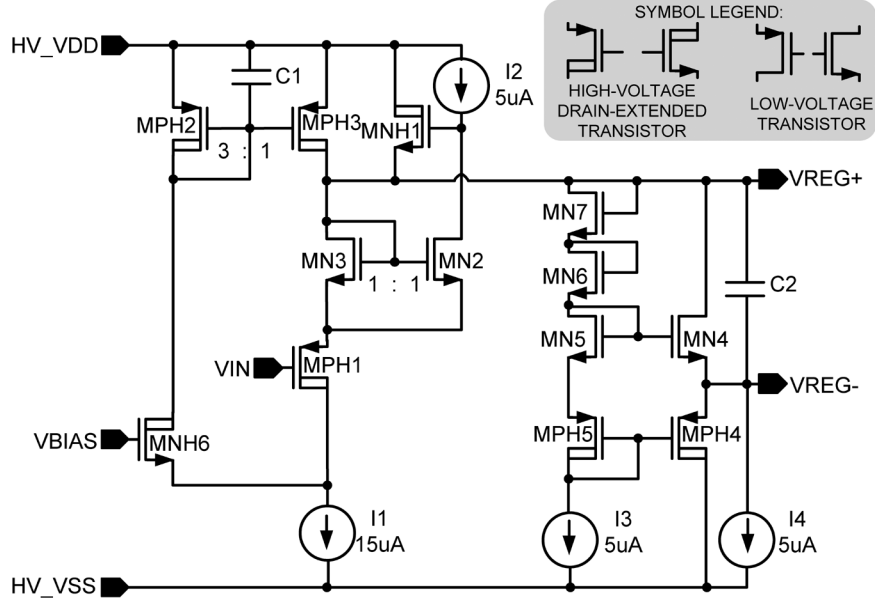


Fig. 7. The floating sub-regulator structure.

and 50 μV . They consume 50 μA and 300 μA respectively, with the output amplifier using the extra current for a robust output drive, current limit and diagnostic functions.

B. Floating Sub-Regulator

As was shown in Fig. 5, a floating sub-regulator referenced to the input voltage is used for each of the input amplifiers. This sub-regulator acts as the supply for the chopping switches and for the clock translator and conditioner. The chosen clock translator requires that the negative supply rail is able to both sink and source current and that its positive rail is able to source current to a switching circuit. Since the chopping switches are also driven from this supply, quick settling of the supply rails is essential to avoid coupling into the signal path.

Fig. 7 shows a simplified schematic of the floating sub-regulator that generates a floating supply of approximately 3 V between rails VREG+ and VREG-, using the input voltage VIN as a reference point. MPH1 and MN3 set the level of the positive rail VREG+ to $2V_{GS}$ above the input voltage, VIN. Throughout most of the input range, VREG+ is regulated by a fast feedback loop formed by MNH1-MN2. Once the common-mode sub-regulator approaches the positive rail, a parallel slower loop formed by MPH1, folded cascode MNH6, and the current mirror MPH2 and MPH3 takes over the control. In this condition MN2

is cut off and thus most of the 15 μA current from the source I1 flows through MNH6, increasing the current in MPH3 threefold. This guarantees operation to the positive rail. The voltage of the negative sub-regulator rail is set to about $3V_{GS}$ below VREG+ by MOS diodes MN5, MN6, MN7 and the buffer MN4, MN5, MPH4, MPH5. Operation to the negative rail is achieved by the addition of the current source I4.

C. Input Stage Configuration

As mentioned previously, high-voltage gate-oxide transistors exhibit significantly higher offset, higher noise and lower gm for comparable area and current levels. Consequently, for best performance, low-voltage transistors must be used in the signal path, especially for the input devices. Fig. 8 shows the arrangement of the input stage of the input amplifiers A_{I1} , A_{I2} . Input transistors MIN1 and MIN2 are cascoded by drain extended devices MC1 and MC2 to protect them. The reference voltage of the cascoding devices follows the common mode voltage, set by the source follower MN and MOS diode MC. This also improves common-mode and power-supply rejection performance, since the drain-source voltage of the input transistors remains nearly constant over the input common-mode and supply range. To further increase the CMRR the tail current source employs impedance boosting. The small OTA A_1

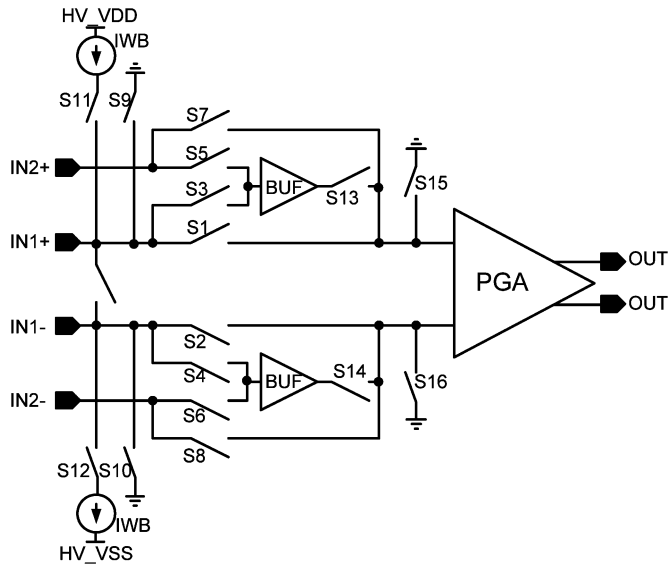


Fig. 9. The input switch network including the HV input current buffer, diagnostic and calibration switches, and wire-break-detection current sources.

slew and settle to the new input value. During the slew process the input protection clamps are activated, increasing the input current and draining charge from the input RC-filter capacitor. Depending on the filter and signal source impedance, the capacitor re-charging can lead to a long settling tail, significantly exceeding the amplifier's own settling time.

To avoid input current and thus improve system-level settling time, an input current buffer, BUF was added as illustrated in Fig. 9. This low precision buffer contains high-voltage MOS transistors with thick 36 V gate-oxide that can tolerate large differential voltages but has degraded offset and noise performance compared to the main amplifier. The buffer is switched into the signal path when an overload condition (MUX channel switch) is anticipated. It supplies the current for the clamps of the precision stage thus preventing current drawn from the input. The duration of the time for which it stays in the signal path is programmable by the user based on system requirements. In Fig. 10 the settling time difference is illustrated when an 8 V input common-mode voltage step is applied to the PGA with and without the input current buffer. The buffer is removed from the signal path after the clamp activating condition disappears and does not affect system accuracy and noise.

C. Input Switch Network

The use of a high voltage CMOS process enables unique functionality not previously integrated on precision industrial instrumentation amplifiers. A typical application system requires calibration and diagnostic modes to measure the input offset and common-mode voltage levels, switch to a system reference source or detect input wire breaks. To accommodate these requirements, the PGA has an integrated input switch network that is shown in Fig. 9. It serves as a 2-channel multiplexer and provides means of reconfiguring the amplifier for system-level trouble-shooting and "housekeeping". For example, the inputs can be shorted for calibration, the single-ended voltage with respect to ground can be measured at either of the inputs or a

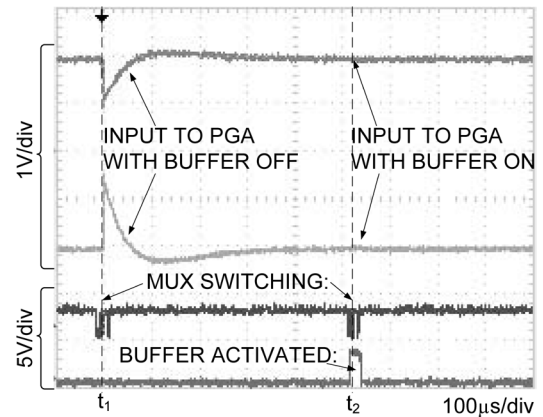


Fig. 10. Settling response to an 8 V step from a voltage source with 10 μA driving capability with and without activating the input current buffer.

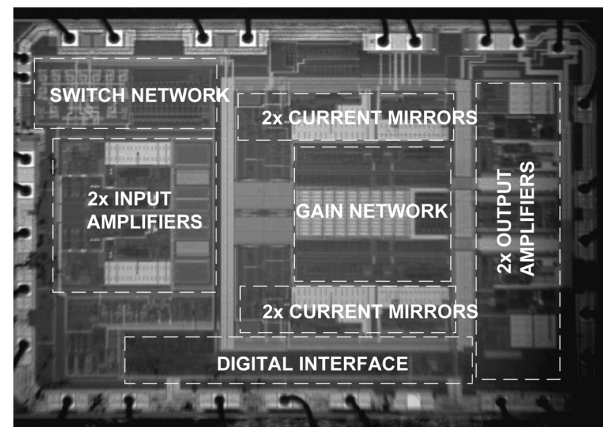


Fig. 11. Chip micrograph.

wire-break at the input can be detected by connecting a small current source to the input pins.

V. MEASUREMENT RESULTS

A chip micrograph of the PGA fabricated in a 0.35 μm analog CMOS process with a 36 V extension is shown in Fig. 11. The chip size is 3.6 $\times$ 2.4 mm. The performance parameters are summarized in Table II. The common-mode rejection ratio (CMRR) distribution of 2000 parts is shown in Fig. 12. The untrimmed CMRR exceeds 120 dB ($< 1 \mu\text{V/V}$ error) at all gain settings. The input offset voltage is better than 20 μV ($G = 128$). The 0.1 Hz to 10 Hz peak-to-peak input referred noise is measured to be 460 nV_{pp} ($G = 128$). Since the noise spectral density is flat due to the chopped architecture, the total noise can be reduced by longer integration. At low gain settings, the input referred offset and noise are higher due to the fact that even though the current mirror provides a signal gain of 1, the noise of its amplifiers is amplified by a factor 5. Despite the V-I and I-V conversions and the multiple supply domains in the signal path, the PGA's gain linearity is better than 0.001% as illustrated in Fig. 13. The PGA has a gain range of 0.125 V/V to 128 V/V in binary steps with an auxiliary gain of 1 or 1.375 implemented around the output amplifiers. The less than 0.1% gain error and less than 2 ppm/ $^{\circ}\text{C}$ gain error drift is determined by the matching of the untrimmed thin-film resistors. The total

TABLE II
SUMMARY OF THE MEASURED PGA PERFORMANCE

<i>Parameter</i>		<i>Mean</i>	<i>Std. dev.</i>
Input Referred Offset Voltage (VOS)	(G=128)	-0.4 μ V	2.3 μ V
	(G=1)	-1.2 μ V*	55 μ V*
VOS Temperature Drift	(G=128)	-0.01 μ V/C	0.06 μ V /C
Gain Non-linearity		0.0003%	0.0002%
Gain Error (GE)		-0.036%	0.022%
GE Temperature Drift		-1.7ppm/C	0.1ppm/C
Common Mode Rejection (CMRR)	(G=128)	-0.03 μ V/V	0.05 μ V/V
	(G=1)	-0.18 μ V/V	0.24 μ V/V
Power Supply Rejection (PSRR)	(G=128)	0.10 μ V/V	0.08 μ V/V
	(G=1)	-0.16 μ V/V	0.73 μ V/V
Supply Current	(High Voltage Input Supplies) (Low Voltage Output Supplies)	2.2mA 0.75mA	
Gain Range		1/8 to 128 in binary steps x (1 or 1.375)	
Input Bias Current		0.25nA	
0.1-10Hz Input Referred Noise	(G=128)	460nVp-p	
	(G=1)	3.8 μ Vp-p*	
Broad-band Noise (Input Referred)	(G=128)	19nVrms/ $\sqrt{\text{Hz}}$	
	(G=1)	200nVrms/ $\sqrt{\text{Hz}}$ *	
Bandwidth	(G $\leq$ 4)	2MHz	
Gain-Bandwidth Product	(G $>$ 4)	8MHz	
CMRR @ 1KHz		90dB	
Chopping Frequency		250KHz	

The statistical data is from 330 samples measured. The drift statistics are from 80 samples measured.

* These specifications are dominated by the four current mirror amplifiers that are in an effective noise gain of 5 to the output.

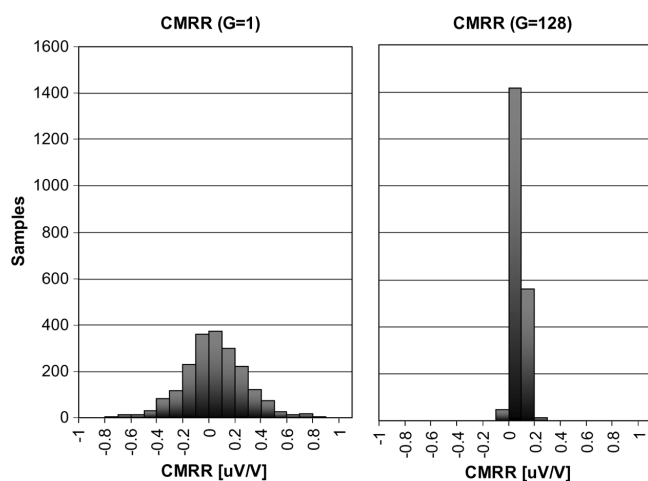


Fig. 12. Common-mode rejection ratio performance of 2000 samples measured in G = 1 and G = 128.

quiescent current is 2.2 mA from the high-voltage input supplies and 750 μ A from the low voltage output supplies. The -3 dB bandwidth in G < 4 is 2 MHz. For gains larger than 4, the -3 dB bandwidth is reduced by half for each 6 dB gain step. Due to the notch filters used in the amplifiers, the chopping glitches are practically invisible even in high gain. This is illustrated on the scope photo taken with a differential probe at the two output voltages and the output referred spectral noise density as is shown in Fig. 14. For this plot, a differential to single-ended converter was used with limited CMRR at the chopping frequency, so the actual differential glitches are even smaller. In

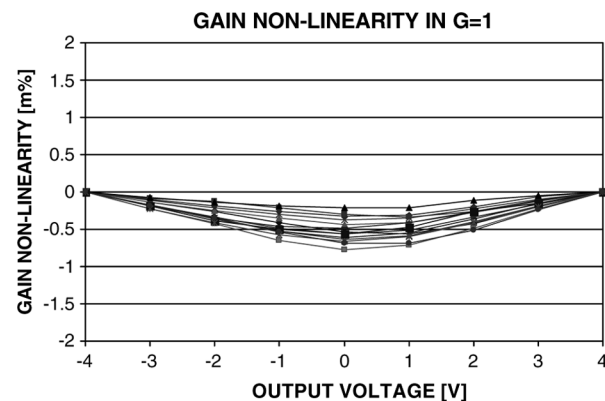


Fig. 13. Measured gain nonlinearity on 15 samples.

Table III, the performance of this PGA is compared to that of other general-purpose HV PGAs.

VI. CONCLUSION

A precision 36 V capable programmable gain amplifier has been presented that is targeted for industrial signal-acquisition applications. By using an architecture in which the input voltage is converted to a current, which is mirrored by precision current mirrors and then converted back into a voltage at the output amplifier, an untrimmed common-mode rejection ratio over 120 dB at all gain settings was achieved. Additionally, this architecture increases the input common-mode range and provides for level translation between the high-voltage input and low voltage output supply domains. The chip is implemented in a 0.35 μ m CMOS process with a 36 V extension. The offset voltage and

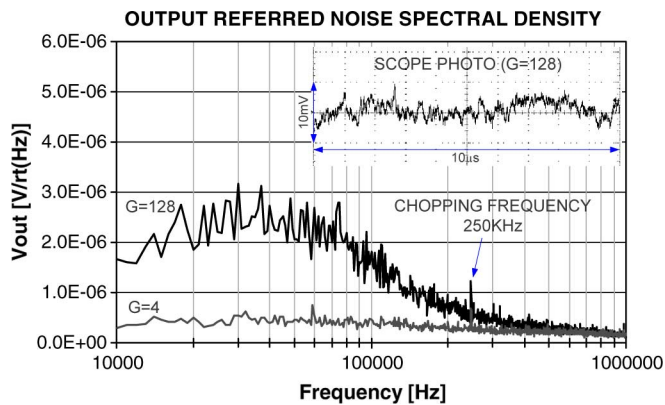


Fig. 14. Output referred spectral plots of the output with $G = 4$ and $G = 128$. The inset illustrates that no chopping glitches are visible on a scope photo of the differential output ($G = 128$).

TABLE III
COMPARISON TO OTHER SIMILAR HIGH-VOLTAGE PGAs

Parameter	THIS PGA	[3]	[4]
Supply Voltage Range	10-36V	9-36V	10-30V
IR Offset Voltage (high gain)	20 μV	62 μV	260 μV
VOS Temperature Drift	0.1 $\mu\text{V}/^\circ\text{C}$	0.6 $\mu\text{V}/^\circ\text{C}$	1.7 $\mu\text{V}/^\circ\text{C}$
Common Mode Rejection Ratio ($G=1$)	120dB	80dB	80dB
On Chip MUX and Diagnostic Functions	yes	no	no

1/f noise were eliminated by the use of chopper stabilization in all internal amplifiers. Despite using chopping throughout the signal path, virtually no glitches are visible at the output due to the utilization of notch-filters inside the amplifiers. In addition to meeting or exceeding the DC performance of existing HV PGAs and IAs, this PGA provides the convenience of a standard digital interface and additional functionality such as provisions for improving system-level settling time, the detection of internal and system-level fault conditions and an input switch network. Apart from acting as a 2-channel multiplexer, this network also enables various system-level diagnostic features.

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